

ARDMORE, OK, USA

WMO: 723529

Lat: 34.147N

Lon: 97.123W

Elev: 844

StdP: 14.25

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 53929

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.4	25.0	6.5	7.9	29.3	10.4	9.6	32.7	18.4	48.8	16.7	50.3	4.1	20	0.409

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	101.5	74.6	98.8	74.0	95.4	73.5	78.7	89.9	78.0	89.6	77.1	88.8	7.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.5	137.6	82.7	74.9	135.2	82.4	73.3	127.9	81.5	42.8	89.9	41.9	90.1	41.0	88.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.9	15.3	12.9	DB	14.1	103.4	6.2	4.3	9.7	106.5	6.1	109.0	2.6	111.5	-1.9	114.6
			WB	12.5	79.3	5.6	2.2	8.5	80.9	5.3	82.2	2.2	83.4	-1.9	85.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.0	43.6	46.8	55.5	63.1	70.8	78.9	83.5	83.5	76.3	64.7	54.6	45.1
	DBStd	16.25	9.71	10.58	9.43	8.12	7.26	4.71	4.63	5.26	6.10	8.40	9.44	8.93
	HDD50	743	243	167	52	8	1	0	0	0	0	6	60	206
	HDD65	2713	663	512	317	127	32	0	0	0	4	110	328	620
	CDD50	5840	46	79	222	402	647	867	1037	788	462	198	53	
	CDD65	2334	1	4	21	71	212	417	572	574	342	101	17	2
	CDH74	26195	15	36	180	576	1892	4424	7289	7213	3480	929	146	15
	CDH80	12690	1	6	29	123	617	1978	3997	4054	1582	280	23	0
Wind	WSAvg	5.3	5.2	6.0	6.6	6.8	6.2	5.5	4.6	4.2	4.3	4.8	5.1	4.9
Precipitation	PrecAvg	35.7	1.7	2.0	2.8	3.3	5.0	4.0	2.3	2.6	4.2	4.0	2.5	2.2
	PrecMax	55.0	6.1	6.5	7.2	13.5	13.4	9.4	6.8	11.7	10.6	16.9	6.7	7.2
	PrecMin	14.8	0.0	0.0	0.4	0.3	0.2	0.6	0.0	0.0	0.1	0.0	0.0	0.1
	PrecStd	8.9	1.5	1.5	1.7	2.6	3.3	2.2	1.9	2.5	2.7	3.3	2.0	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.4	79.1	82.5	88.3	93.3	99.0	106.1	106.3	99.5	90.6	82.4	75.3
		MCWB	58.3	60.7	63.2	66.8	73.7	75.6	73.9	73.8	71.2	69.0	65.0	60.8
	2%	DB	70.3	72.6	79.1	82.4	90.2	95.0	100.3	102.2	94.8	86.5	78.9	69.9
		MCWB	55.9	57.5	62.3	66.5	73.2	74.0	74.6	74.1	71.8	68.8	62.6	59.0
	5%	DB	64.3	68.4	74.7	80.6	86.3	91.3	98.5	99.3	91.0	82.2	73.2	64.3
		MCWB	53.5	57.2	60.9	65.4	71.0	74.4	74.1	74.3	71.8	66.6	61.1	55.3
	10%	DB	61.1	63.8	71.7	77.1	82.3	90.0	95.1	96.8	89.7	79.3	70.3	61.1
		MCWB	49.9	54.8	59.1	63.4	70.7	73.9	73.8	73.8	71.1	65.5	60.3	52.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.6	65.8	67.8	74.2	78.5	80.2	79.8	79.4	78.9	74.2	69.5	67.1
		MCDB	68.9	71.1	76.8	81.3	87.3	89.8	91.9	90.9	86.5	83.0	76.3	70.6
	2%	WB	60.8	62.8	65.7	71.1	76.3	78.4	78.5	78.2	77.1	72.2	67.1	62.4
		MCDB	66.2	68.0	73.7	77.3	84.5	88.6	90.9	91.1	85.9	81.1	74.3	66.8
	5%	WB	55.2	59.7	63.8	69.0	74.6	77.4	77.8	77.3	75.3	70.6	64.3	58.3
		MCDB	63.1	66.7	70.8	75.6	82.7	87.4	90.3	90.9	84.1	78.4	70.0	62.7
	10%	WB	50.7	55.6	61.8	67.1	72.9	76.1	76.9	76.3	73.9	68.8	61.6	53.1
		MCDB	59.6	63.7	69.0	74.6	81.0	85.3	89.2	89.5	83.1	75.8	67.7	58.8
Mean Daily Temperature Range	5% DB	MDBR	21.2	20.6	20.8	21.3	19.3	18.5	20.4	21.5	21.3	22.0	21.4	19.9
		MCDBR	29.1	27.1	25.0	23.5	22.4	20.6	24.7	25.2	24.2	25.1	25.2	26.4
		MCWBR	17.2	16.3	12.8	12.1	9.4	7.0	6.1	6.2	7.8	10.9	13.2	16.8
	5% WB	MCDBR	24.5	23.1	19.8	18.8	18.5	17.6	19.9	20.9	18.8	18.7	20.4	20.4
		MCWBR	17.6	17.5	12.5	12.3	9.9	7.5	6.7	6.4	7.9	9.9	14.1	17.0
Clear-Sky Solar Irradiance	taub	0.304	0.312	0.350	0.380	0.416	0.429	0.432	0.432	0.391	0.349	0.325	0.306	
	taud	2.500	2.492	2.408	2.344	2.310	2.312	2.340	2.310	2.380	2.485	2.490	2.496	
	Ebn at Noon	289	298	293	288	277	272	271	269	276	281	278	281	
	Edh at Noon	27	30	35	39	41	41	40	40	36	30	27	25	
All-Sky Solar Radiation	RadAvg	892	1066	1407	1722	1848	2094	2134	1951	1621	1252	948	770	
	RadStd	93	161	96	137	191	209	128	115	156	182	121	92	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page